

| testcoverage (pygeodesy 26.8.16): 95%                  |            |         |          |          |
|--------------------------------------------------------|------------|---------|----------|----------|
| coverage.py v7.14.0, created at 2026-08-14 12:30 -0400 |            |         |          |          |
| File                                                   | statements | missing | excluded | coverage |
| pygeodesy/__init__.py                                  | 193        | 4       | 8        | 98%      |
| pygeodesy/albers.py                                    | 408        | 13      | 30       | 97%      |
| pygeodesy/angles.py                                    | 494        | 23      | 27       | 95%      |
| pygeodesy/auxilats/_CX_4.py                            | 6          | 0       | 0        | 100%     |
| pygeodesy/auxilats/_CX_6.py                            | 6          | 0       | 0        | 100%     |
| pygeodesy/auxilats/_CX_8.py                            | 6          | 0       | 0        | 100%     |
| pygeodesy/auxilats/_CX_Rs.py                           | 84         | 3       | 59       | 96%      |
| pygeodesy/auxilats/__init__.py                         | 8          | 0       | 0        | 100%     |
| pygeodesy/auxilats/auxAngle.py                         | 224        | 9       | 33       | 96%      |
| pygeodesy/auxilats/auxDLat.py                          | 149        | 7       | 10       | 95%      |
| pygeodesy/auxilats/auxDST.py                           | 98         | 3       | 35       | 97%      |
| pygeodesy/auxilats/auxLat.py                           | 437        | 31      | 47       | 93%      |
| pygeodesy/auxilats/auxily.py                           | 108        | 6       | 37       | 94%      |
| pygeodesy/azimuthal.py                                 | 318        | 7       | 55       | 98%      |
| pygeodesy/basics.py                                    | 279        | 38      | 171      | 86%      |
| pygeodesy/booleans.py                                  | 966        | 51      | 145      | 95%      |
| pygeodesy/cartesianBase.py                             | 322        | 27      | 42       | 92%      |
| pygeodesy/clipy.py                                     | 316        | 14      | 18       | 96%      |
| pygeodesy/constants.py                                 | 225        | 6       | 82       | 97%      |
| pygeodesy/css.py                                       | 234        | 6       | 12       | 97%      |
| pygeodesy/datums.py                                    | 297        | 20      | 9        | 93%      |
| pygeodesy/deprecated/__init__.py                       | 13         | 0       | 0        | 100%     |
| pygeodesy/deprecated/bases.py                          | 5          | 0       | 3        | 100%     |
| pygeodesy/deprecated/classes.py                        | 72         | 2       | 203      | 97%      |
| pygeodesy/deprecated/consterns.py                      | 20         | 0       | 28       | 100%     |
| pygeodesy/deprecated/datum.py                          | 7          | 0       | 0        | 100%     |
| pygeodesy/deprecated/functions.py                      | 42         | 2       | 237      | 95%      |
| pygeodesy/deprecated/nvector.py                        | 6          | 0       | 6        | 100%     |
| pygeodesy/deprecated/rhumbBase.py                      | 3          | 0       | 0        | 100%     |
| pygeodesy/deprecated/rhumbaux.py                       | 4          | 0       | 0        | 100%     |
| pygeodesy/deprecated/rhumbsolve.py                     | 4          | 0       | 0        | 100%     |
| pygeodesy/deprecated/rhumbx.py                         | 4          | 0       | 0        | 100%     |
| pygeodesy/dms.py                                       | 298        | 10      | 41       | 97%      |
| pygeodesy/ecef.py                                      | 508        | 27      | 43       | 95%      |
| pygeodesy/ecefLocals.py                                | 49         | 4       | 2        | 92%      |
| pygeodesy/elevations.py                                | 69         | 14      | 51       | 80%      |
| pygeodesy/ellipses.py                                  | 401        | 28      | 27       | 93%      |
| pygeodesy/ellipsoidalBase.py                           | 260        | 15      | 64       | 94%      |
| pygeodesy/ellipsoidalBaseDI.py                         | 331        | 31      | 25       | 91%      |
| pygeodesy/ellipsoidalExact.py                          | 42         | 0       | 1        | 100%     |
| pygeodesy/ellipsoidalGeodSolve.py                      | 39         | 0       | 1        | 100%     |
| pygeodesy/ellipsoidalKarney.py                         | 40         | 0       | 6        | 100%     |
| pygeodesy/ellipsoidalNvector.py                        | 134        | 5       | 5        | 96%      |
| pygeodesy/ellipsoidalVincenty.py                       | 178        | 2       | 11       | 99%      |
| pygeodesy/ellipsoids.py                                | 766        | 29      | 78       | 96%      |
| pygeodesy/elliptic.py                                  | 564        | 18      | 63       | 97%      |
| pygeodesy/epsg.py                                      | 100        | 2       | 7        | 98%      |
| pygeodesy/errors.py                                    | 332        | 31      | 59       | 91%      |
| pygeodesy/etm.py                                       | 410        | 34      | 115      | 92%      |
| pygeodesy/fmath.py                                     | 354        | 38      | 124      | 89%      |
| pygeodesy/formy.py                                     | 449        | 11      | 25       | 98%      |
| pygeodesy/frechet.py                                   | 261        | 11      | 78       | 96%      |
| pygeodesy/fstats.py                                    | 325        | 6       | 24       | 98%      |
| pygeodesy/fsums.py                                     | 1107       | 54      | 176      | 95%      |
| pygeodesy/gars.py                                      | 144        | 5       | 10       | 97%      |
| pygeodesy/geod3solve.py                                | 144        | 13      | 30       | 91%      |
| pygeodesy/geodesici.py                                 | 617        | 55      | 373      | 91%      |
| pygeodesy/geodesicw.py                                 | 233        | 22      | 26       | 91%      |
| pygeodesy/geodesicx/_C4_24.py                          | 6          | 0       | 0        | 100%     |
| pygeodesy/geodesicx/_C4_27.py                          | 6          | 0       | 0        | 100%     |
| pygeodesy/geodesicx/_C4_30.py                          | 6          | 0       | 0        | 100%     |
| pygeodesy/geodesicx/__init__.py                        | 6          | 0       | 0        | 100%     |
| pygeodesy/geodesicx/gx.py                              | 523        | 39      | 32       | 93%      |
| pygeodesy/geodesicx/gxarea.py                          | 230        | 16      | 34       | 93%      |
| pygeodesy/geodesicx/gxbases.py                         | 60         | 4       | 1        | 93%      |
| pygeodesy/geodesicx/gxline.py                          | 256        | 22      | 29       | 91%      |
| pygeodesy/geodsolve.py                                 | 86         | 2       | 120      | 98%      |
| pygeodesy/geohash.py                                   | 368        | 10      | 42       | 97%      |
| pygeodesy/geoids.py                                    | 688        | 26      | 122      | 96%      |
| pygeodesy/hausdorff.py                                 | 235        | 11      | 57       | 95%      |
| pygeodesy/heights.py                                   | 316        | 15      | 122      | 95%      |
| pygeodesy/internals.py                                 | 275        | 25      | 119      | 91%      |
| pygeodesy/interns.py                                   | 353        | 2       | 19       | 99%      |
| pygeodesy/iters.py                                     | 204        | 7       | 27       | 97%      |
| pygeodesy/karney.py                                    | 362        | 30      | 70       | 92%      |
| pygeodesy/ktm.py                                       | 233        | 11      | 39       | 95%      |
| pygeodesy/latlonBase.py                                | 475        | 34      | 86       | 93%      |
| pygeodesy/lazily.py                                    | 216        | 9       | 83       | 96%      |
| pygeodesy/lcc.py                                       | 288        | 10      | 25       | 97%      |
| pygeodesy/ltp.py                                       | 331        | 17      | 33       | 95%      |
| pygeodesy/ltpTuples.py                                 | 503        | 18      | 49       | 96%      |
| pygeodesy/mgrs.py                                      | 267        | 2       | 62       | 99%      |
| pygeodesy/named.py                                     | 530        | 21      | 147      | 96%      |
| pygeodesy/namedTuples.py                               | 366        | 32      | 2        | 91%      |
| pygeodesy/nvectorBase.py                               | 256        | 13      | 33       | 95%      |
| pygeodesy/osgr.py                                      | 305        | 8       | 65       | 97%      |
| pygeodesy/points.py                                    | 530        | 36      | 100      | 93%      |
| pygeodesy/props.py                                     | 243        | 7       | 64       | 97%      |
| pygeodesy/resections.py                                | 375        | 11      | 60       | 97%      |
| pygeodesy/rhumb/__init__.py                            | 10         | 0       | 0        | 100%     |
| pygeodesy/rhumb/aux_.py                                | 140        | 6       | 2        | 96%      |
| pygeodesy/rhumb/bases.py                               | 373        | 24      | 116      | 94%      |
| pygeodesy/rhumb/ekx.py                                 | 225        | 4       | 46       | 98%      |
| pygeodesy/rhumb/solve.py                               | 96         | 8       | 67       | 92%      |
| pygeodesy/simplify.py                                  | 210        | 2       | 3        | 99%      |
| pygeodesy/solveBase.py                                 | 247        | 22      | 40       | 91%      |
| pygeodesy/sphericalBase.py                             | 250        | 14      | 11       | 94%      |
| pygeodesy/sphericalNvector.py                          | 317        | 28      | 38       | 91%      |
| pygeodesy/sphericalTrigonometry.py                     | 388        | 26      | 71       | 93%      |
| pygeodesy/streprs.py                                   | 273        | 15      | 19       | 95%      |
| pygeodesy/trf.py                                       | 622        | 28      | 62       | 95%      |
| pygeodesy/triaxials/__init__.py                        | 10         | 0       | 0        | 100%     |
| pygeodesy/triaxials/bases.py                           | 516        | 44      | 16       | 91%      |
| pygeodesy/triaxials/conformal3.py                      | 272        | 34      | 26       | 88%      |
| pygeodesy/triaxials/triaxial3.py                       | 442        | 44      | 27       | 90%      |
| pygeodesy/triaxials/triaxial5.py                       | 429        | 44      | 39       | 90%      |
| pygeodesy/units.py                                     | 305        | 13      | 12       | 96%      |
| pygeodesy/unitsBase.py                                 | 122        | 7       | 9        | 94%      |
| pygeodesy/ups.py                                       | 161        | 5       | 10       | 97%      |
| pygeodesy/utily.py                                     | 342        | 20      | 47       | 94%      |
| pygeodesy/utm.py                                       | 259        | 8       | 21       | 97%      |
| pygeodesy/utmups.py                                    | 87         | 2       | 13       | 98%      |
| pygeodesy/utmupsBase.py                                | 269        | 5       | 29       | 98%      |
| pygeodesy/vector2d.py                                  | 344        | 8       | 25       | 98%      |
| pygeodesy/vector3d.py                                  | 235        | 8       | 44       | 97%      |
| pygeodesy/vector3dBase.py                              | 322        | 38      | 165      | 88%      |
| pygeodesy/webmercator.py                               | 126        | 1       | 18       | 99%      |
| pygeodesy/wgrs.py                                      | 190        | 6       | 9        | 97%      |
| Total                                                  | 29993      | 1639    | 5389     | 95%      |
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